Feasibility Study

Feasibility is defined as the practical extent to which a project can be performed successfully. A feasibility study determines whether the solution proposed for accomplishing the requirements is practical and workable. It includes considerations such as resource availability, development costs, expected benefits, and maintenance expenses. The outcome should guide whether to proceed with requirements engineering and system development.

If a system does not support the business objectives, it holds no real value to the organization. Many systems fail due to unclear goals, undefined requirements, or organizational issues influencing procurement. Therefore, the objective of this feasibility study is to ensure that TravelEase Hub is acceptable to users, adaptable to change, and aligned with recognized standards.

Various other objectives of feasibility study are listed below:

- To analyse whether the TravelEase Hub will meet organizational requirements
- To determine if it can be implemented using current technologies within a reasonable budget and schedule.
- To evaluate whether it can be integrated with existing systems and operations.

Types of Feasibility

Various types of feasibility that are commonly considered include technical feasibility, operational feasibility, and economic feasibility.

Technical feasibility

Technical feasibility assesses the current resources (such as hardware and software) and technology, which are required to accomplish user requirements in the software within the allocated time and budget. Technical feasibility also performs the following tasks:

- Analyses the technical skills and capabilities of the software development team members.
- Determines whether the relevant technology is stable and established.

- Ascertains that the technology chosen for software development has a large number of users so that they can be consulted when problems arise or improvements are required.

In the “TravelEase Hub” project, the technical feasibility is strongly supported by the use of a React.js, Node.js, and Express.js stack with Supabase (a PostgreSQL-based backend-as-a-service), which are stable and widely adopted technologies. The backend system, built with Node.js and Express.js, ensures scalable operations, while React.js delivers a dynamic and responsive frontend. The use of AI and ML technologies like TensorFlow.js and the OpenAI API helps to generate intelligent features such as itinerary planning and budget estimation. The system also utilizes Google Maps API for map integration and JWT for secure user authentication. Hosting is managed via Vercel, enabling fast and serverless deployment. These technology choices ensure the system can be developed efficiently using well-established tools with strong community support.

Operational feasibility

Operational feasibility assesses the extent to which the required software performs a series of steps to solve business problems and user requirements. This feasibility is dependent on human resources (software development team) and involves visualizing whether the software will operate after it is developed and be operative once it is installed. Operational feasibility also performs the following tasks:

- Determines whether the problems anticipated in user requirements are of high priority.
- Determines whether the solution suggested by the software development team is acceptable.
- Analyses whether users will adapt to new software.
- Determines whether the organization is satisfied by the alternative solutions proposed by the software development team.

In “TravelEase Hub”, operational feasibility is achieved through user-friendly features, including a clean UI/UX, dashboards for both users and agencies, and real-time updates. The system provides personalized travel planning, AI-based suggestions, booking management, and group coordination tools such as pooling and expense splitting. Travel agencies benefit from simplified package management and real-time booking alerts. The onboarding process is straightforward, and the interface is designed to minimize the learning curve. As the platform aligns with user behaviour and modern expectations, users and agencies are likely to adapt quickly and integrate the system into their operations.

Economic feasibility

Economic feasibility determines whether the required software is capable of generating financial gains for an organization. It involves the cost incurred on the software development team, estimated cost of hardware and software, cost of performing feasibility study, and so on.

- Cost incurred on software development to produce long-term gains for an organization.
- Cost required conducting a full software investigation (such as requirements elicitation and requirements analysis).
- Cost of hardware, software, development team, and training.

The "TravelEase Hub" project is economically feasible, as it is developed using open-source technologies that reduce licensing and infrastructure costs. It eliminates the need for physical hardware by relying on cloud-based deployment. Revenue is generated through bookings and optional agency registrations. The platform offers essential features for free, increasing accessibility and encouraging adoption. Its economic model supports sustainability by combining low development costs with high user value and engagement. As the user base grows, the system is expected to produce long-term revenue through expanded services making the project financially beneficial for stakeholders.